

Surjyadip Sen

From embedded systems to atmospheric AI: Building high-efficiency solutions for complex domains

📍 Chennai, India ✉ surjyadip100.sen@gmail.com ☎ +91 9330415099 🌐 <https://surjyadipsen.in> 🐙 [Github](#) 🔗 [LinkedIn](#)

SUMMARY

Full Stack Developer & Atmospheric Data Scientist specializing in distributed deep learning, weather forecasting modeling, and scalable real-time data analytics. Proficient in Python, JAX, Flax NNX, PyTorch, C++, and Next.js, with experience in training and fine-tuning state-of-the-art weather models (Pangu, Graphcast, Gencast) on HPC/distributed systems, and designing high-performance IoT and web architectures.

EXPERIENCE

India Meteorological Department New Delhi, India
Intern (Atmospheric Data Scientist) May 2026 - Jun 2026
Trained/fine-tuned weather forecasting models (Pangu, Graphcast, Gencast, FGN, IMDGFS) using distributed computing. Conducted extensive comparative analysis of results against state-of-the-art baselines.

Tech Stack: Python, JAX, CUDA, Flax NNX, Pytorch, Numpy, Scipy

Integrity Technologies Pvt Ltd Remote
Intern (Web and App Developer) Jun 2025 - Feb 2026
Built/maintained website and mobile app, migrated tech stacks, integrated payment gateways, improved developer experience, and managed infrastructure to reduce system failures.

Tech Stack: React Native, Expo, Django, React, Next.js, Jenkins, Celery, PostgreSQL, AWS

PROJECTS

Raspberry Pi 5 Homelab May 2023 - Jun 2026
Built a containerized Raspberry Pi 5 homelab with NVMe SSD boot storage. Configured Docker Compose services (via SSH), Traefik reverse proxy (dynamic routing/SSL), and Tailscale for portless remote access to Immich and Paperless-ngx.

Tech Stack: Docker Compose, Traefik, Tailscale, Linux, Raspberry Pi 5, NVMe SSD, Bash, SSH

EdgeAI Hand Gesture Classifier Aug 2025 - Mar 2026
Developed an autonomous hand gesture classifier using KNN with SIMD acceleration on ESP32-S3. Achieved microsecond-range inference (98% accuracy) and on-device training via SPIFFS.

Tech Stack: C, C++, ESP-IDF, FreeRTOS, ESP32-S3, TinyML, SIMD, Xtensa LX7, SPIFFS

CryptoDataAggregator Nov 2023 - Jul, 2025
Optimized multi-exchange crypto trading via unified databases. Built pipelines for custom timeframes, indicator calculations, portfolio analysis, and real-time alerts/entry signals.

Tech Stack: Python, Postgresql, Celery, Redis, Docker (compose)

SKILLS

 Languages Python, C/C++, JavaScript, TypeScript	 Frameworks & Libraries JAX, Flax NNX, PyTorch, SciPy, NumPy, Pandas, Django, FastAPI, React, Next.js, React Native, Expo, Tailwind CSS
 Tools & Systems Git, Bun, Node, Docker, Linux, Traefik, Nginx, Jenkins, PBS (Portable Batch System), CUDA	

EDUCATION

Vellore Institute of Technology Computer Science	B.Tech CSE Core Chennai, India • 2025-2029
Kendriya Vidyalaya Barrackpore (Afs) Science (CS)	Higher Secondary • 87% Barrackpore, West Bengal, India • 2013-2025